الأكاديمية العربية للعلوم والتكنولوجيا والنقل البحري
Arab Academy for Science, Technology & Maritime Transport
College of Computing and Information Technology

CAI3103/CS460

DEEP LEARNING

LECTURE-6

DL Hyperparameters

Introduction to Hyperparameters in ML and DL

What are Hyperparameters?

- **Definition:** External configurations set by the practitioner before the training process begins.
- **Hyperparameters vs. Parameters:**
  - **Parameters:** Learned automatically from data during training (e.g., weights w, biases b).
  - **Hyperparameters:** Manually tuned to control the learning process (e.g., learning rate, number of layers).
- **Goal:** To find the optimal balance between underfitting and overfitting (Generalization).

Examples of ML algorithms Hyperparameters

Objective: Hierarchical partitioning of data based on feature-value splits.

Parameter	Definition	Effect of High Value
max_depth	Max nodes from root to leaf.	High complexity; likely to overfit.
min_samples_split	Min samples to split a node.	Prevents small, specific leaves.
Criterion	Split quality measure.	Uses Gini Impurity or Entropy (Information Gain).

Decision Tree

Objective: A feedforward neural network designed for complex non-linear mapping.

Parameter	Definition	Selection Logic
Hidden Layers	Neurons per layer.	Increases model capacity for complexity.
Activation	Non-linear transform.	ReLU (standard), Sigmoid, or Tanh.
Solver	Optimization algorithm.	Adam (adaptive) or SGD (stochastic).

MLP ANN

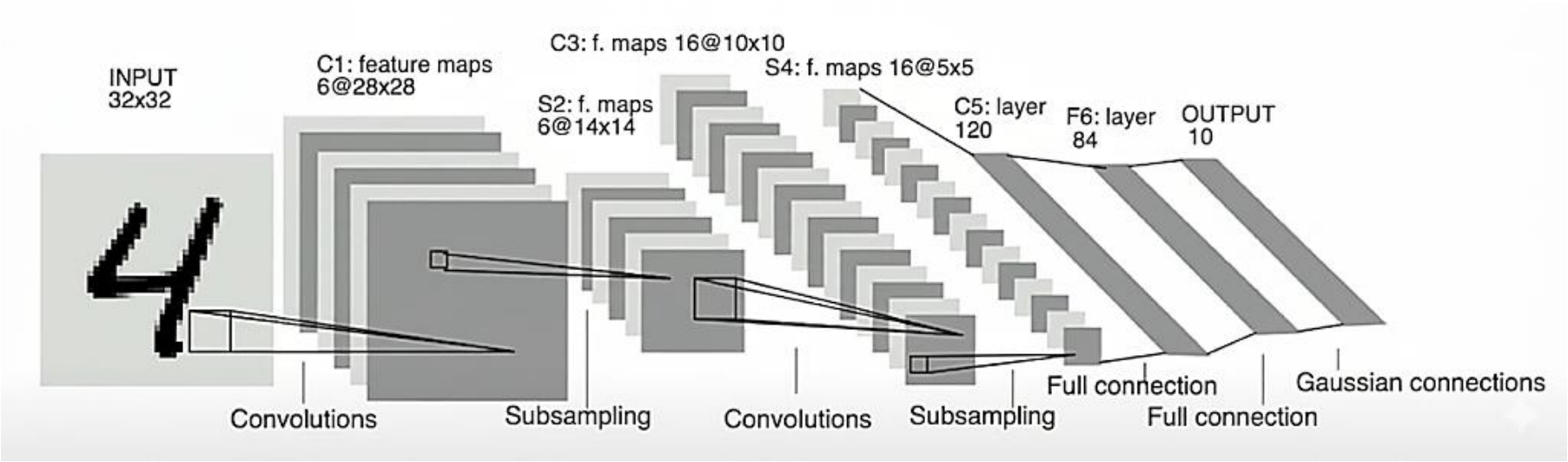
Convolutional Neural Networks (CNNs) DL Hyperparameters

- **Common DL Hyperparameters:**
  - **Learning Rate:** Step size during gradient descent.
  - **Batch Size:** Number of samples processed before updating weights.
  - **Epochs:** Number of full passes through the dataset.
  - **Dropout:** Percentage of neurons randomly disabled to prevent overfitting.

Structure of **LeNet-5** Convolutional Neural Network (CNN)

The hyperparameters within LeNet-5 structure are divided into:

- The spatial dimensions → the **height and width** of the input data (images)
- The number of filters → the **depth** of each hidden layer (representing the number of unique features the model can learn at that stage)
- The connectivity between layers → describes **how neurons in one layer link to the next** (Sparse/Full Connectivity)



Hyperparameters of **LeNet-5** Convolutional Neural Network (CNN)

1. Input Layer

- **Dimensions:** 32×32 pixels.
- **Type:** Grayscale (1 channel).

2. Layer C1 (Convolutional Layer)

- **Feature Maps:** 6.
- **Kernel Size:** 5×5.
- **Stride:** 1 (inferred by the output size 28×28).
- **Padding:** 0 (Valid padding).
- **Trainable Parameters:** $(5 \times 5 \times 1 + 1) \times 6 = 156$.

3. Layer S2 (Subsampling/Pooling Layer)

- **Type:** Average Pooling.
- **Feature Maps:** 6.
- **Kernel Size:** 2×2.
- **Stride:** 2.
- **Output Dimensions:** 14×14.

4. Layer C3 (Convolutional Layer)

- **Feature Maps:** 16.
- **Kernel Size:** 5×5.
- **Stride:** 1.
- **Output Dimensions:** 10×10.

5. Layer S4 (Subsampling/Pooling Layer)

- **Type:** Average Pooling.
- **Feature Maps:** 16.
- **Kernel Size:** 2×2.
- **Stride:** 2.
- **Output Dimensions:** 5×5.

6. Layer C5 (Fully Connected Convolutional Layer)

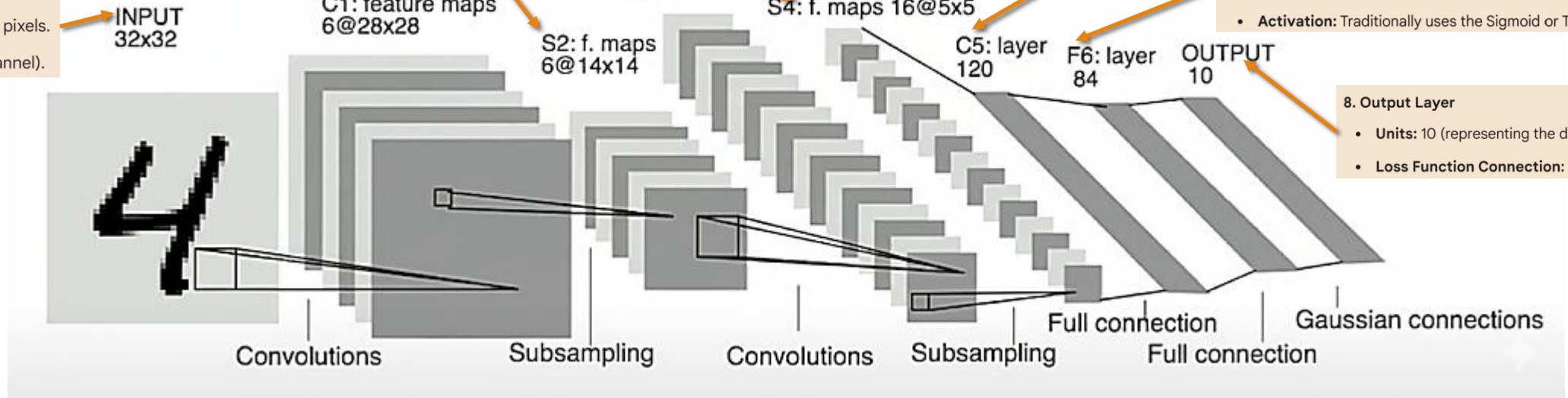
- **Units:** 120.
- **Kernel Size:** 5×5.

7. Layer F6 (Fully Connected Layer)

- **Units:** 84.
- **Activation:** Traditionally uses the Sigmoid or Tanh activation function

8. Output Layer

- **Units:** 10 (representing the digits 0-9).
- **Loss Function Connection:** Radial Basis Function (RBF)



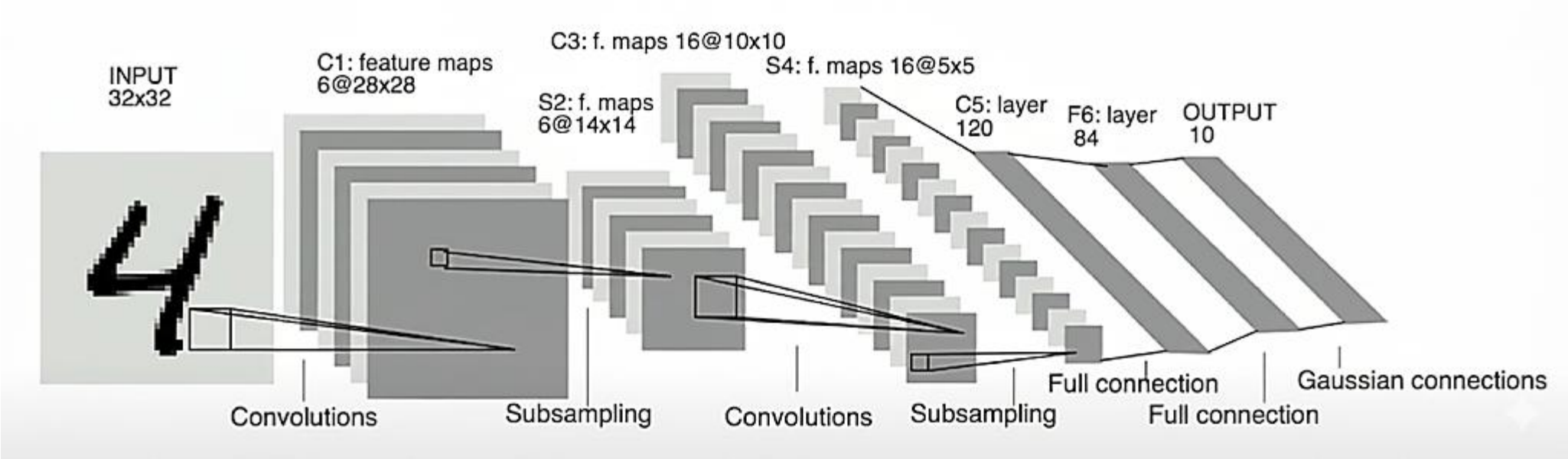
Subsampling (The LeNet-5 Approach) vs Pooling

1. **Pooling (Max Pooling or Average Pooling):**
   - It is a fixed operation with no trainable parameters.
   - **Max Pooling**: Selects the maximum value from a neighborhood (set of pixels in the feature map).
   - **Average Pooling**: Calculates the arithmetic mean of the neighborhood.

   → <u>It is a purely mathematical transformation with No weights or biases involved.</u>

2. **Subsampling (The LeNet-5 Approach):**
   - It is a specific form of average pooling, defined by the original LeNet-5 authors, that includes trainable parameters.
   - It takes a 2X2 neighborhood, averages the four pixel values, multiplies the result by a trainable weight, and adds a trainable bias, then the result is passed through a non-linear activation function (sigmoid).

   → <u>Unlike conventional pooling, the network can learn how much importance to give to each pooled region through the considered weights and biases.</u>

Connectivity between layers in **LeNet-5** CNN

- **Sparse Connectivity**: In layer C3, each feature map is only connected to a specific subset of the 6 maps in S2. This reduces the parameter count and forces the filters to learn complementary information.
- **Full Connectivity**: In the final layers (C5, F6), every neuron is connected to every neuron in the previous layer to aggregate all extracted features for classification.